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EX NO: 2b

CALCULATION AND VERIFICATION OF FREQUENCY
DEVIATION AND MODULATION INDEX OF FM SIGNALS
USING MATLAB SIMULATION

Objectives:

To generate a Frequency Modulated (FM) signal in MATLAB using specified message and carrier signal parameters, and to determine the frequency deviation and modulation index of the

FM signal while analyzing the relationship between frequency deviation, message frequency, and

modulation index.

PROGRAM (MATLAB CODE)

% Calculation and Verification of Frequency Deviation

% and Modulation Index of FM Signal

clc;

clear;

close all;

%% Message Signal Parameters

$A_m = 2$;

% Message amplitude (V)

$f_m = 500$;

% Message frequency (Hz)

%% Carrier Signal Parameters

$A_c = 1$;

% Carrier amplitude (V)

$f_c = 10000$; % Carrier frequency (Hz)

%% FM Parameters

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kf = 250;

% Frequency sensitivity (Hz/V)

fs = 100000; % Sampling frequency (Hz)

T = 0.01;

% Simulation duration (s)

%% Time Vector

t = 0:1/fs:T;

%% Message Signal

m = Am*cos(2*pi*fm*t);

%% Frequency Deviation

delta_f = kf*Am;

%% Modulation Index

beta = delta_f/fm;

%% FM Signal

fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

%% Display Parameters

fprintf('Carrier Frequency (fc) = %.0f Hz\n',fc);

fprintf('Message Frequency (fm) = %.0f Hz\n',fm);

fprintf('Message Amplitude (Am) = %.0f V\n',Am);

fprintf('Frequency Sensitivity (kf) = %.0f Hz/V\n',kf);

fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n',delta_f);

fprintf('Modulation Index (Beta) = %.2f\n',beta);

%% Verification

beta_verify = delta_f/fm;

fprintf('Verified Modulation Index = %.2f\n',beta_verify);

%% Verification Check

tolerance = 1e-6;

if abs(beta - beta_verify) < tolerance

fprintf('\nVerification Successful\n');

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else

fprintf('\nVerification Failed\n');

end

%%% Plotting

figure;

subplot(2,1,1);

plot(t,m,'LineWidth',1.5);

grid on;

title('Message Signal');

xlabel('Time (s)');

ylabel('Amplitude (V)');

subplot(2,1,2);

plot(t,fm_signal,'LineWidth',1);

grid on;

title('FM Signal');

xlabel('Time (s)');

ylabel('Amplitude (V)');

```

OUTPUT:

